Efficient Global Content Recovery for Multi-View Data

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April 16, 2026

Multi-view Data with Partial Observability

Latent Factor Decomposition:

- **Observation ($\mathbf{x}_k$):** Data generated from content and style.
- **Style ($\mathbf{s}_k$):** View-specific factors (e.g., lighting).
- **Content ($\mathbf{c}$):** Shared invariant factors (e.g., identity).

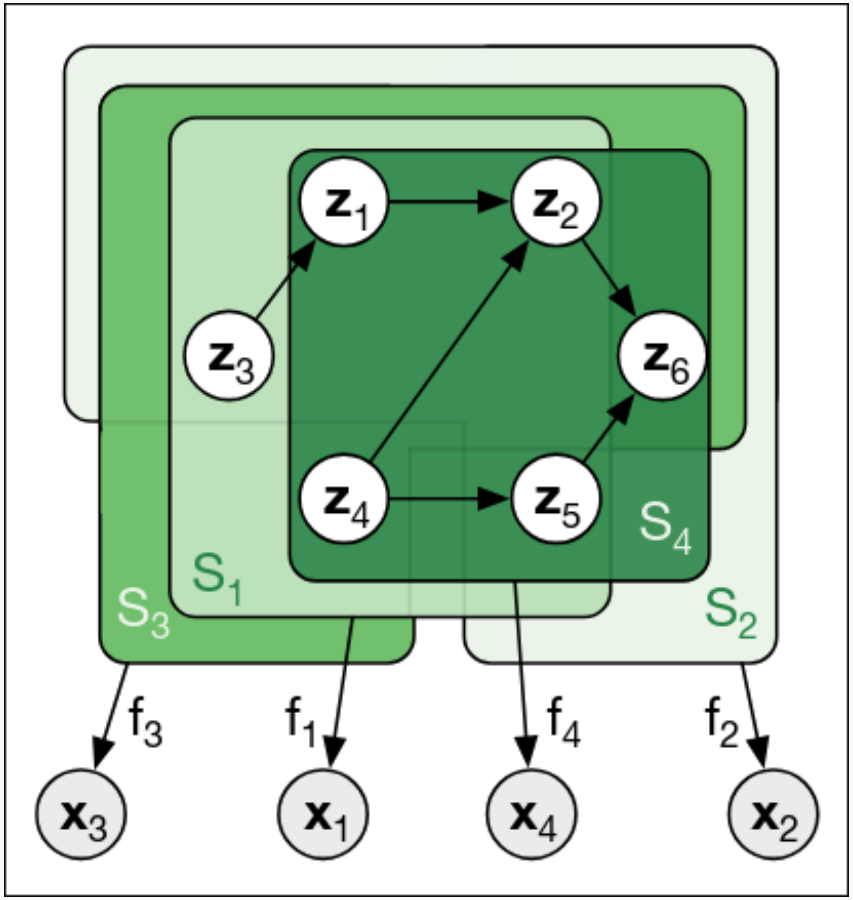

Figure 1: Multi-view Data Setup. Source: Yao et al. (2024) [1].

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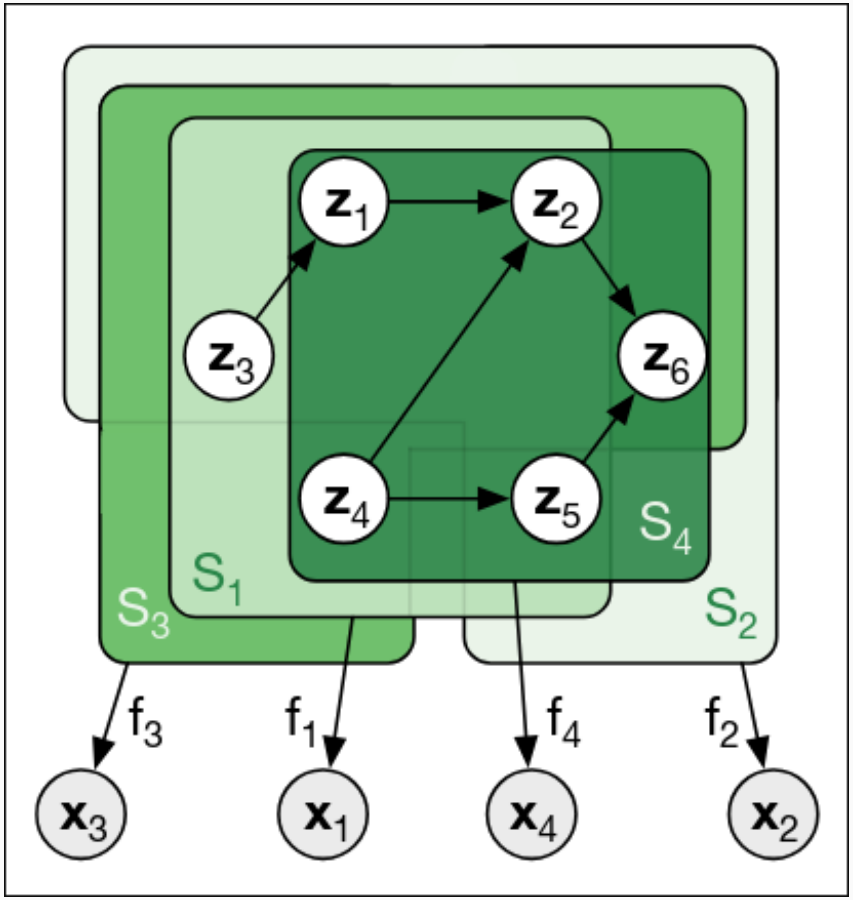
Goal: Identify $\mathbf{c}$ despite only observing partial subsets of the environment.



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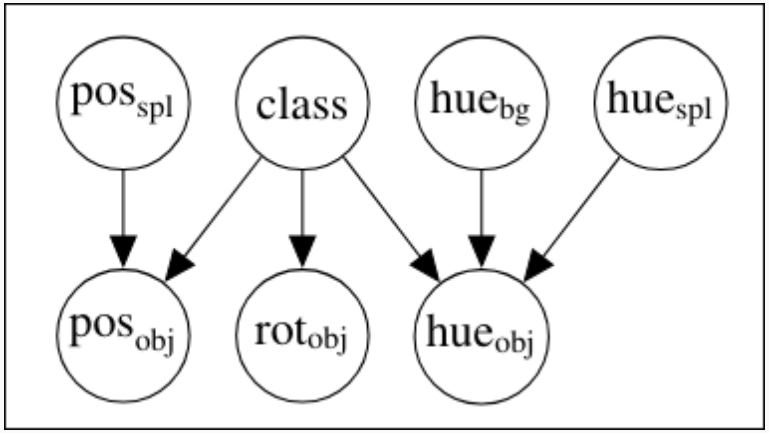
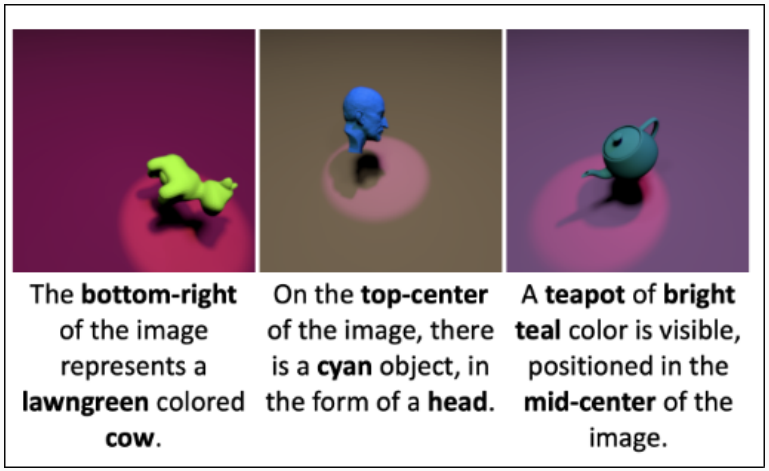
Example: Multimodal3DIdent



(a) Causal3DIdent: causal graph illustrating the data generation process [2].



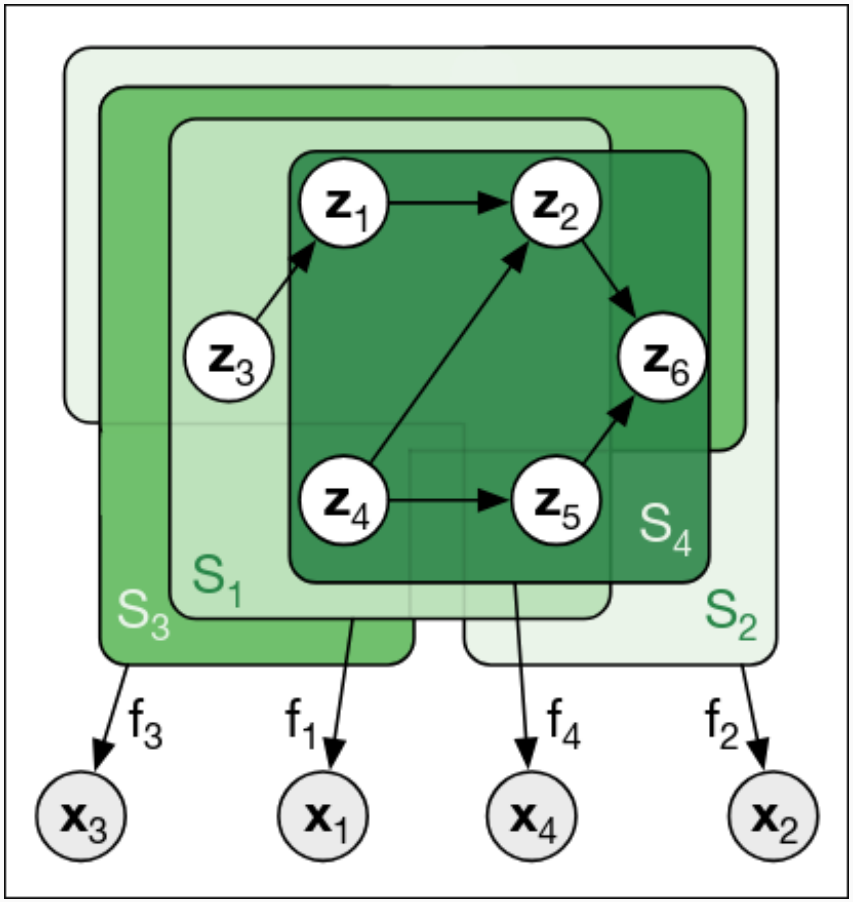
(b) Multimodal3DIdent: Examples of image-caption pairs from the dataset. Each view captures different subsets of latent factors [3].

Figure 2: Example of multi-view data with partial observability.

The Challenge of Identifiability

The Ideal: Full Identifiability

- Recover *every* latent factor individually ($c_1, c_2, \ldots$).
- Mapping $\hat{\mathbf{c}}$ to $\mathbf{c}$ must be trivial (permutation/scaling).



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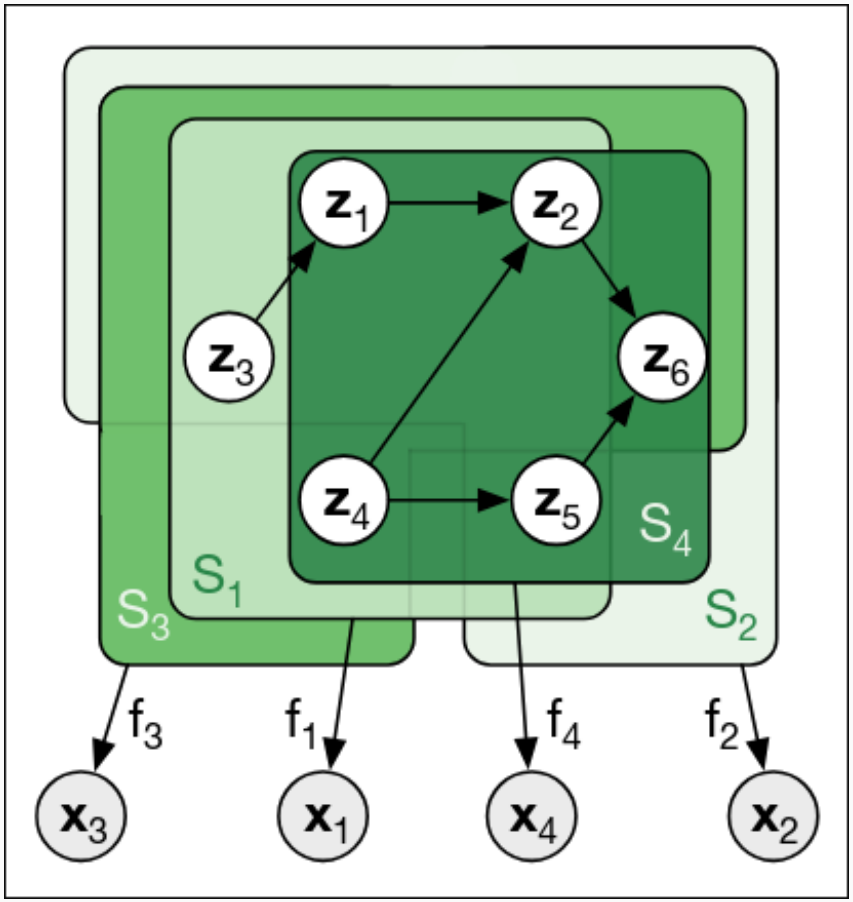
The Reality:

- Requires latent variables to be independent and restrictive assumptions on the mixing functions.
- Unrealistic for real-world data.



Relaxation: Block-Identifiability

Definition

The true content $\mathbf{c}$ is **block-identified** [2] by an encoder $g(\mathbf{x})$ if there exists a smooth invertible mapping h such that:

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Key Properties

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Why "Block"?

- We recover the joint information as a "block" but it may be entangled.
- Sufficient for downstream causal tasks and prediction.

Contrastive Learning

Contrastive Learning: Maximize agreement between representations of different views of the same content while minimizing agreement between different contents.

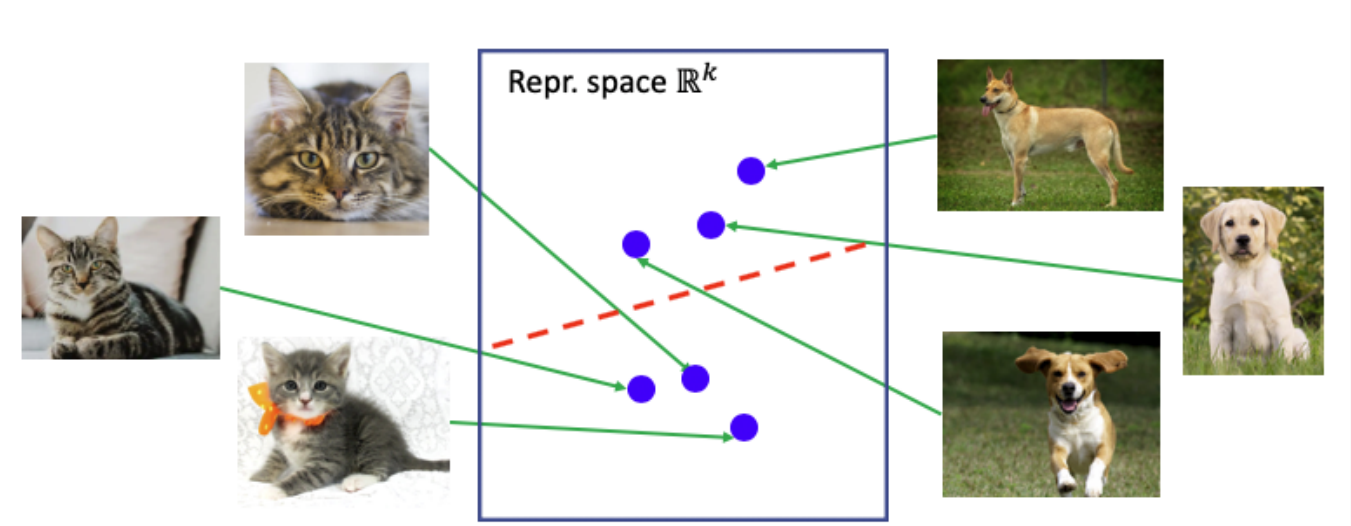

Figure 3: Contrastive learning example. Pictures of dogs (and cats) should be close in representation space, but dogs and cats should be far apart.

Block-Identification via Contrastive Learning

Yao et al. (2024) [1] show that, under certain assumptions, contrastive learning can achieve block-identifiability of the content $\mathbf{c}$ in multi-view data by minimizing the contrastive loss:

$$\mathcal{L} = \underbrace{\sum_{k<k' \in V} \mathbb{E}\left[\|g_k(\mathbf{x}_k) - g_{k'}(\mathbf{x}_{k'})\|_2\right]}_{\text{Content Alignment}} - \underbrace{\sum_{k \in V} H(g_k(\mathbf{x}_k))}_{\text{Entropy regularization}} ,$$

where $g_k : \mathcal{X}_k \to \mathbb{R}^{|\mathbf{c}|}$ is the encoder for view k, and $H(\cdot)$ denotes the differential entropy.

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The Known Content Dimension Constraint

The dimensionality of the encoder output **must exactly match** the true content dimension: $\dim(\hat{\mathbf{c}}) = \dim(\mathbf{c})$.

The Known Content Dimension Constraint

This is a very strong requirement that is often violated in practice:

- **Underestimation:** If $\dim(\hat{\mathbf{c}}) < \dim(\mathbf{c})$, we lose information and cannot recover the full content.
- **Overestimation:** If $\dim(\hat{\mathbf{c}}) > \dim(\mathbf{c})$, the encoder may capture spurious style information, leading to entangled representations and loss of block-identifiability.

A masking approach

Yao et al. (2024) [1] show that for view-specific encoders $R = \{r_k\}$, content selectors $\Phi = \{\phi_V^k\}$, and projections $T = \{t_k\}$ solving:

$$\min \mathrm{Reg}(\Phi) := \sum_{V \in \mathcal{V}} \sum_{k \in V} \|\phi_V^k\|_0 \quad \text{subject to:} \quad R, \Phi, T \in \arg\min \mathcal{L}(R, \Phi, T)$$

where $\mathrm{Reg}(\Phi)$ is an L_0 sparsity penalty on the selectors, and the base loss is:

$$\mathcal{L}(R, \Phi, T) = \sum_{V \in \mathcal{V}} \sum_{k < k' \in V} \mathbb{E}\left[\|\phi_V^k \oslash r_k(\mathbf{x}_k) - \phi_V^{k'} \oslash r_{k'}(\mathbf{x}_{k'})\|_2\right] - \sum_{k \in V} H(t_k \circ r_k(\mathbf{x}_k))$$

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Known View-Specific Latent Dimension Constraints

The view-specific encoders are assumed to map to the exact dimension matching the true dimension of the subset of latents generating the view.

Literature Overview: Handling Unknown $|c|$

1. **Heuristic Search** *(Traditional)*

   Iterative training with different $|\hat{\mathbf{c}}|$ values.

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3. **Sparsity-Inducing Penalties** *(Modern)*

   Using ℓ_0 regularization to prune dimensions during training.

   - **Drawback:** Non-differentiable and computationally expensive.

Our Proposed Method

We introduce a *learnable shared global gate* $\phi \in [0,1]^{\dim(\hat{c})}$ applied to the output of all view encoders, by means of a hard concrete distribution [4]:

- **Stochastic Binary Mask**
- **Continuous relaxation of ℓ_0**
- **Differentiable end-to-end training**

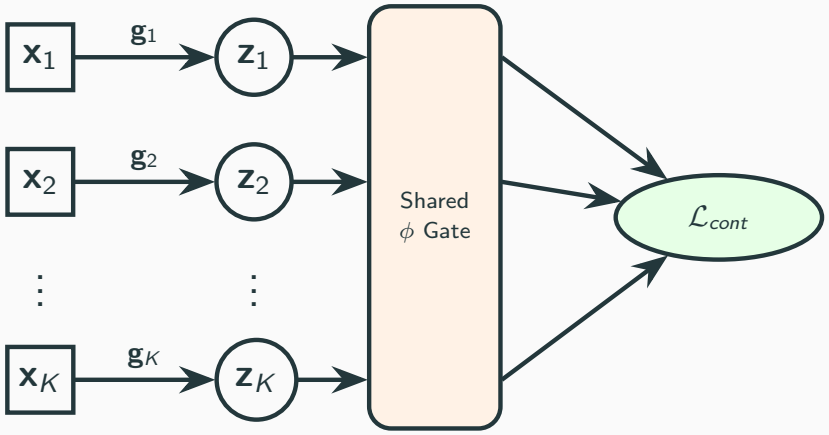
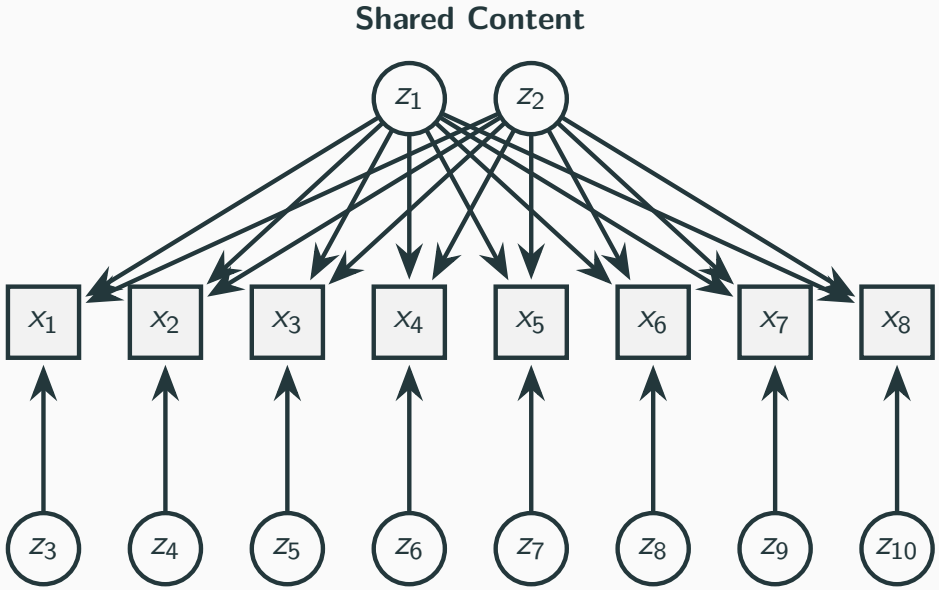


Figure 4: Global Gating Architecture

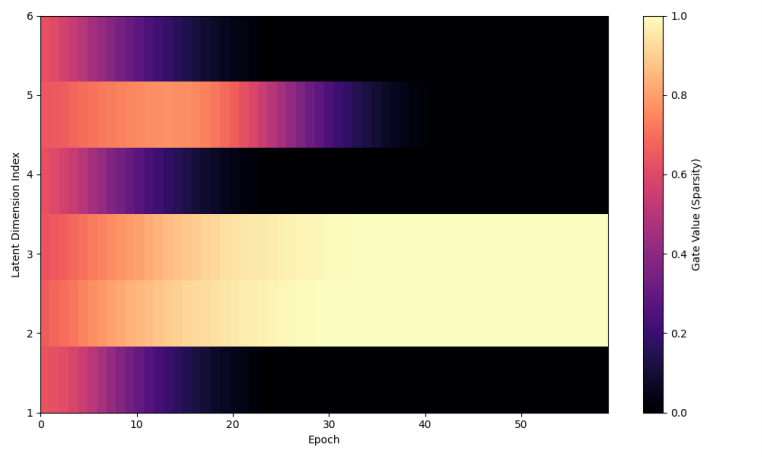
Numerical Experiment: Independent Factors

Setup Details:

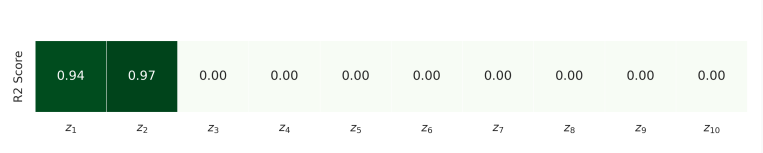
- 8 views, 10 total latents.
- **Gaussian latents:** $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
- $\mathbf{z_1}, \mathbf{z}_2$ are shared content.
- $\mathbf{z}_3, \ldots, \mathbf{z}_{10}$ are view-specific style.
- Overestimate content dimension: $\dim(\hat{\mathbf{c}}) = 6$.



Numerical Results - Independent Latent Factors



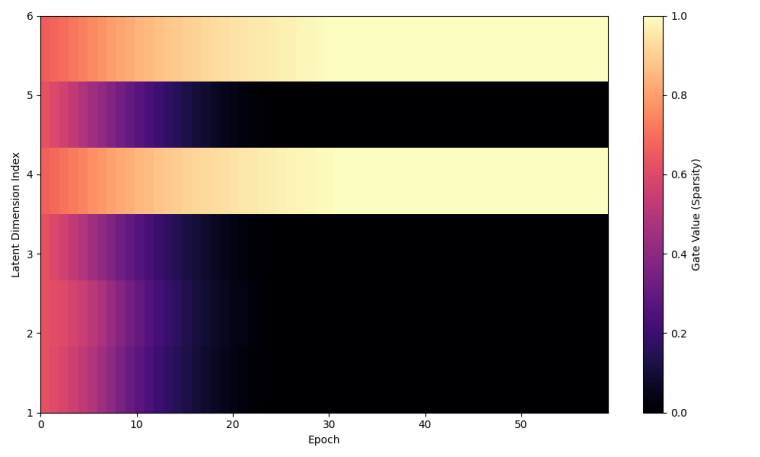
(a) Gate convergence history



(b) R^2 performance per latent factor.

Figure 5: Numerical Experiment Results (**independent** latent factors).

Numerical Results - Dependent Latent Factors



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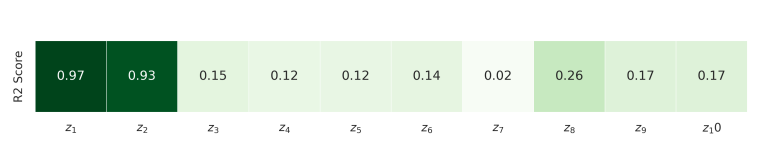
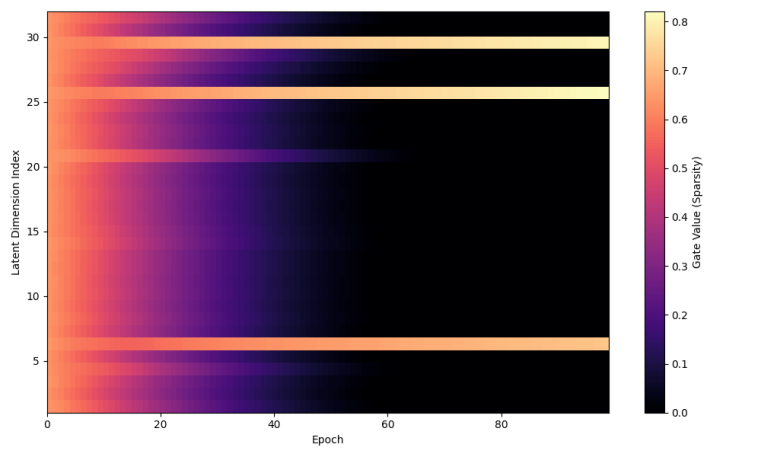


(b) R^2 performance per latent factor.

Figure 6: Numerical Experiment Results (**dependent** latent factors).

Multimodal3DIdent Results



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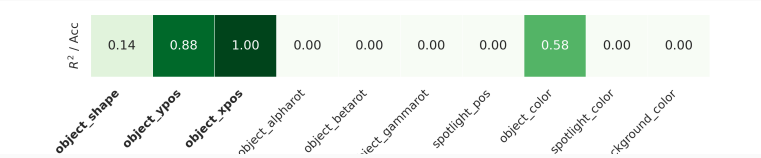

(b) Prediction performance per latent factor.

Figure 7: Multimodal Experiments on Multimodal3DIdent Dataset.

Discussion

Key Takeaways:

- Our global gating method successfully identifies the correct number of content dimensions in both synthetic and real multimodal datasets.
- The method is robust to overestimation of the content dimension, consistently recovering the shared content structure without significant entanglement with style.

Limitations & Future Work

Lack of Formal Proofs: While empirical results are strong, the theoretical guarantees for block-identifiability under the *relaxed overestimation* setting are yet to be established.

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Future Research Directions

- **Theoretical Conditions:** Establish formal requirements for block-identifiability when content dimensions are overestimated.
- **Sensitivity Analysis:** Investigate how the *degree* of overestimation affects training convergence/cost.

Conclusion

Summary:

- We introduced a method for block-identifiable content recovery in multi-view data with partial observability, where the content dimension is unknown and potentially overestimated.
- Our approach leverages a shared global gate to learn a sparse representation during training.
- Empirical results on synthetic and real datasets demonstrate that the true number of content dimensions are matched, with comparable prediction performances.

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